

Question				Solution	Marks
3	(a)			Taking moments about the handle, By principle of moments Sum of anticlockwise moments = sum of clockwise moments $W(1.10) + R \cos 73^\circ (0.90) = R \sin 73^\circ (1.10 + 0.50)$ $R = \frac{100(1.10)}{1.60 \sin 73^\circ - 0.90 \cos 73^\circ} = 86.82$ $R = 87 \text{ N}$	M1 A1
	(b)			Since net force = 0, Horizontal component $H_x = R \cos 73^\circ = 25.38 \text{ N}$ Vertical component $H_y = 100 - R \sin 73^\circ = 16.97 \text{ N}$ $\tan \theta = \frac{H_y}{H_x} = \frac{16.97}{25.38}$ $\theta = 34^\circ$ $H = \sqrt{H_x^2 + H_y^2} = 30.53 \text{ N}$ $H = 31 \text{ N}$ at an angle of 34° above the horizontal (towards the left)	M1 M1 A1

Question			Solution	Marks
4	(a)		Consider the body moving with initial velocity u with a constant	